

Question			Marking details				Marks available													
							AO1	AO2	AO3	Total	Maths	Prac								
6	(a)	(i)	35 [m]					1		1	1									
		(ii)	60 [km/h]					1		1	1									
		(iii)	80 [km/h]					1		1	1									
		(iv)	[Alert driver – thinking distance =] 25 [m] (1) [Tired driver – doubles to] 50 [m] (1) [Braking distance of 20 m stays the same giving stopping distance of] 70 [m] (1) so disagree To award 3 marks the conclusion must be present						3	3										
		(v)	25 seen anywhere (1) 17 seen anywhere (1) Time = 1.47 [s] (1)					3		3	3									
	(b)		<table><tr><td>Thinking distance</td><td>Braking distance</td><td>Stopping distance</td><td>Impact speed</td></tr><tr><td>Stays the same (1)</td><td>Increases (1)</td><td>Increases</td><td>Increases (1)</td></tr></table>				Thinking distance	Braking distance	Stopping distance	Impact speed	Stays the same (1)	Increases (1)	Increases	Increases (1)	2		1	3		
Thinking distance	Braking distance	Stopping distance	Impact speed																	
Stays the same (1)	Increases (1)	Increases	Increases (1)																	

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	(c)		<table><tr><td>Action</td><td>Seat belt</td><td>Crumple zone</td></tr><tr><td>Increases the time of the collision</td><td></td><td>✓</td></tr><tr><td>Reduces force on the car</td><td></td><td>✓ (1)</td></tr><tr><td>Prevents driver continuing through the windscreen</td><td>✓ (1)</td><td></td></tr></table>	Action	Seat belt	Crumple zone	Increases the time of the collision		✓	Reduces force on the car		✓ (1)	Prevents driver continuing through the windscreen	✓ (1)		2			2		
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Ignore any ticks in row 1 More than 1 tick in a row no marks can be awarded																					
			Question 6 total	4	6	4	14	6	0												